



SQL DATA Analytics

Project



SQL Projects



Data Warehousing

"Organize, Structure, Prepare,"

- ETL/ELT Processing
- Data Architecture
- Data Integration
- Data Cleansing
- Data Load
- Data Modeling



Exploratory Data Analysis (EDA)

"Understand Data,"

- Basic Queries
- Data Profiling
- Simple Aggregations
- Subquery



Advanced Data Analytics

"Answer Business Questions,"

- Complex Queries
- Window Functions
- CTE
- Subqueries
- Reports

Dimensions vs Measures

Dataset

~	~	~	~	~
~	~	~	~	~
~	~	~	~	~
~	~	~	~	~

Is it Numeric ?

& Does it make Sense to aggregate?

No

Yes



Dimension

Category

Products

Birthdate

ID



Measure

Sales

Quantity

Age

A	C
B	D

Dimensions Exploration

DISTINCT [Dimension]

DISTINCT Country

DISTINCT Category

DISTINCT Product

$$\begin{array}{c}
 \vdots \\
 \hline A \\
 \hline B \\
 \hline B \\
 \hline C \\
 \hline A \\
 \hline B \\
 \hline C \\
 \hline A \\
 \hline \vdots
 \end{array}
 \Rightarrow
 \frac{A}{\frac{B}{C}}$$



Date Exploration

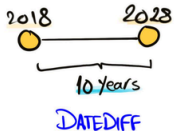
MIN/MAX [Date Dimension]

MIN Order_date

MAX Create_date

MIN Birthdate

2019
2020
2018
2018
2022
2023
2023
2023
2022



999

Measures Exploration

Σ [Measure]

SUM (Sales)

AVG (Price)

SUM (Quantity)

10
20
50
30
10
80
30
10

$\Rightarrow$

BIG Number

240

Key Metric



Magnitude

Σ [Measure] By [Dimension]

Total Sales By Country

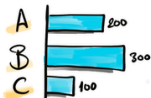
Total Quantity By Category

Average Price By Product

Total Orders By Customer

600

A	200
B	300
C	100





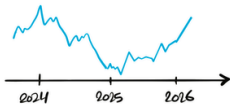
Change-Over-Time Trends~

Σ [Measure] By [Date Dimension]

Total Sales By Year

Average Cost By Month

2024	300
2025	100
2026	200





Cumulative Analysis

Σ [Cumulative Measure] By [Date Dimension]

Running Total Sales By Year

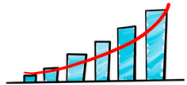
Moving Average of Sales By Month

Cumulative

2024	300	300
2025	100	400
2026	200	600

Diagram illustrating cumulative analysis with a table and arrows showing the accumulation of values over time.

WINDOW FUNCTIONS





Performance Analysis

Current [Measure] - Target [Measure]

Current Sales - Average Sales

Current Year Sales - Previous Year Sales

Current Sales - lowest Sales

Diagram illustrating Performance Analysis:

- Current (indicated by a blue arrow pointing to the first column)
- Target (Avg) (indicated by a red circle with a minus sign and a blue arrow pointing to the second column)
- Performance (indicated by a blue arrow pointing to the third column)

A	200	- 200	0
B	300	- 200	100
C	100	- 200	-100



WINDOW FUNCTIONS



Part-to-Whole

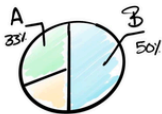
Proportional
Analysis

$(\text{[Measure]} / \text{Total [Measure]}) * 100$ By [Dimension]

$(\text{Sales} / \text{Total Sales}) * 100$ By Category

$(\text{Quantity} / \text{Total Quantity}) * 100$ By Country

A	200	33%
B	300	50%
C	100	17%





Data Segmentation

[Measure] By [Measure]

Total Products By Sales Range

Total Customers By Age

Σ → Categorize

3	50		
4	100	→	Low
5	150	→	Medium
1	200	→	Large
10	250		
5	300		
			7
			6
			15

CASE WHEN STATEMENT

